MSIN0097 Predictive Analytics — Individual Coursework

Predicting Household Purchasing Power from Spending Patterns

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Word count: 1,993
Github link:

1. Problem Framing

Economic policy rests on knowing who is poor, yet this knowledge is surprisingly difficult to obtain. In 2023, over 13 million people in the United Kingdom lived in relative poverty, and an estimated 40 per cent of eligible households did not claim means-tested support (DWP, 2024; Policy in Practice, 2022). If income is hard to observe directly, spending is not: every household leaves a fingerprint in its expenditure. This project exploits that insight, asking whether purchasing power can be inferred from what households spend rather than what they earn.

The analysis draws on the Living Costs and Food Survey (LCFS) for 2021 to 2023, a nationally representative diary survey covering 14,294 households across the post-COVID recovery and the cost-of-living crisis (ONS and DEFRA, 2025). Rather than using raw income, the target variable is equivalised income: gross household income divided by the OECD modified equivalence scale (1.0 for the first adult, +0.5 per additional adult, +0.3 per child under 14), thereby adjusting for household size and composition (OECD, 2013). This reverses the conventional direction of economic analysis: instead of predicting spending from income, the model infers income status from spending.

The task is formalised as multi-class classification into five equivalised income quintiles (Q1 = lowest purchasing power through Q5 = highest), evaluated by macro F1-score to ensure all quintiles receive equal weight. All income-derived variables are excluded to prevent target leakage. This framing carries direct policy implications. A household that declares Q1 income to HMRC while exhibiting expenditure patterns characteristic of Q5 presents a discrepancy warranting investigation for potential tax non-compliance. More broadly, the ability to infer purchasing power from spending alone enables proactive welfare targeting and real-time cost-of-living monitoring without requiring income disclosure.

Throughout this project, Claude Code was employed as an agent tool to scaffold the code structure, suggest model architectures, and generate baseline pipelines. Every output was verified for leakage, statistical soundness, and domain correctness. The Appendix documents what was accepted, modified, or rejected and why.

2. Exploratory Data Analysis

The merged dataset presents 1,915 columns, a feature space in which target leakage poses a significant risk. Rather than hand-picking features, Pearson correlations were computed between every column and equivalised income. This analysis identified 227 variables that encode income directly or indirectly: the p300–p499 range (income aggregates and tax deductions, with p344p reaching r = 0.999), all b-prefix columns (means-tested benefits whose amounts are determined by income), and several administrative codes. After exclusion, 52 clean features remained: 27 expenditure categories and 25 demographic indicators, each selected with a documented correlation coefficient.

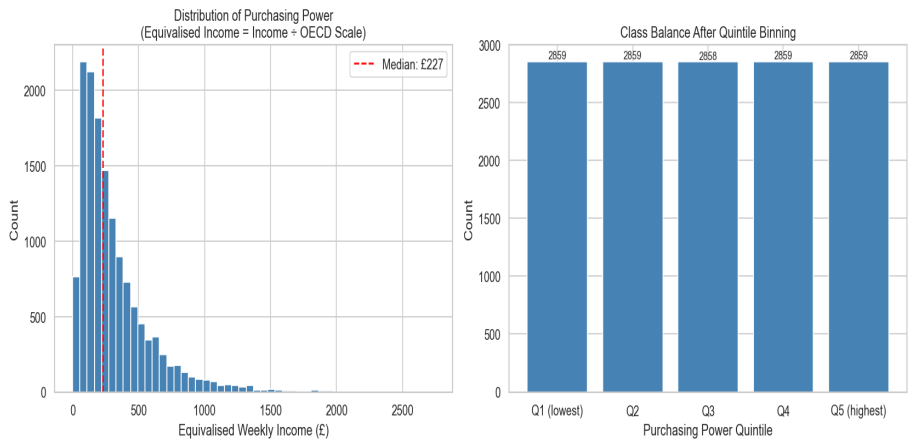

Figure 1: Left panel shows right-skewed equivalised income distribution. Right panel shows balanced quintile classes

Equivalised income has a mean of £311.93 per week and a median of £226.78 per week, exhibiting pronounced right skew (range £0 to £2,740). The OECD scale spans 1.0 (single adult) to 6.5 (large family), implying that raw income can overstate the purchasing power of larger households by a factor of up to 6.5. Quintile binning yields balanced classes of approximately 2,859 households each, thereby avoiding the class imbalance that would arise from fixed income thresholds.

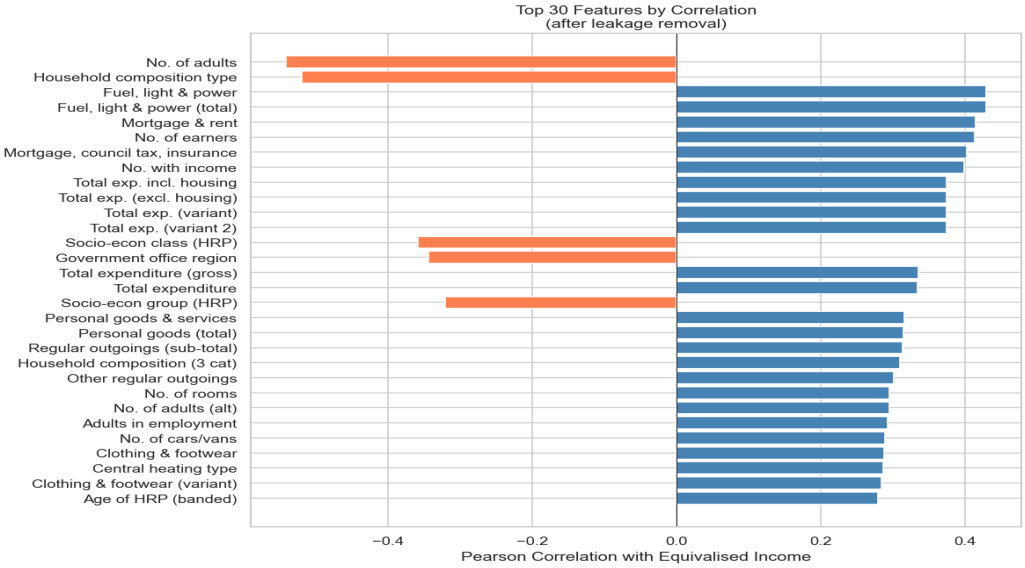

Figure 2: Systematic feature selection — correlation with equivalised income across all non-leakage columns.

Beneath these clean distributions, the data concealed quality problems invisible to standard diagnostics. Standard null-value detection reported zero per cent missing, yet deeper inspection revealed empty strings masquerading as valid entries, 90 negative expenditure values across five columns (likely refunds signalling prior large purchases), and high zero rates in mortgage payments (88.1%), public transport (89.5%), and council tax (64.1%). Importantly, these zeros represent valid economic signals rather than missing data; a zero mortgage payment, for instance, identifies an outright homeowner. Empty strings were converted to NaN for imputation; negatives were retained.

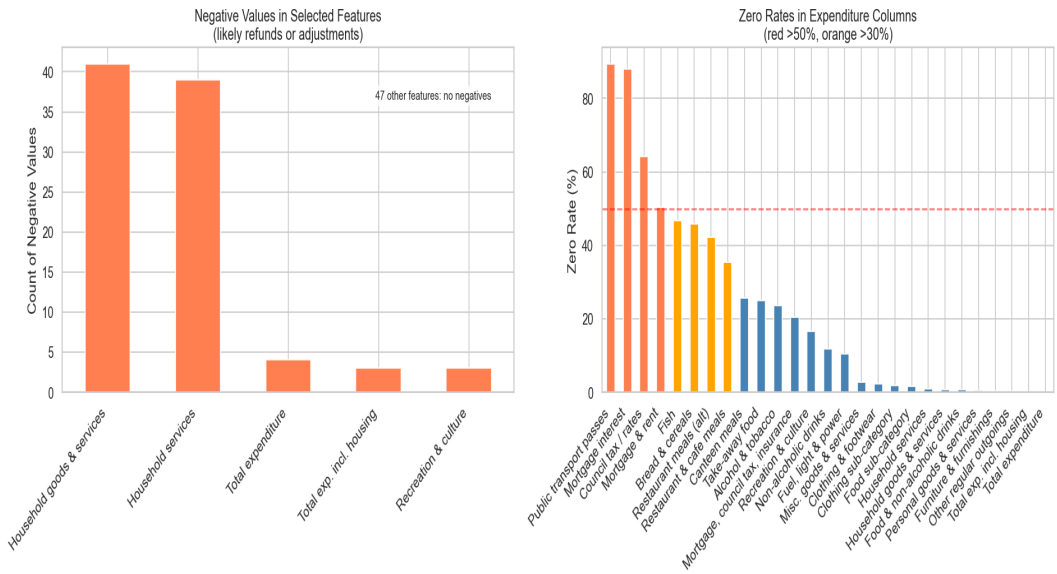

Figure 3: Effective missingness (left) and zero rates (right) revealing hidden data quality issues.

Perhaps the most analytically significant finding concerns how spending differentiates income tiers. Q5/Q1 ratios reveal that mortgage interest varies 200-fold between the richest and poorest quintiles, mortgage and rent 29-fold, and regular outgoings 16-fold. Food, by contrast, differs only 1.8-fold, consistent with Engel's Law: all households require food, yet lower-income households allocate a disproportionately larger share of expenditure to this category (Engel, 1857). These patterns confirm the central premise: spending encodes purchasing power, and the signal is strongest in discretionary categories.

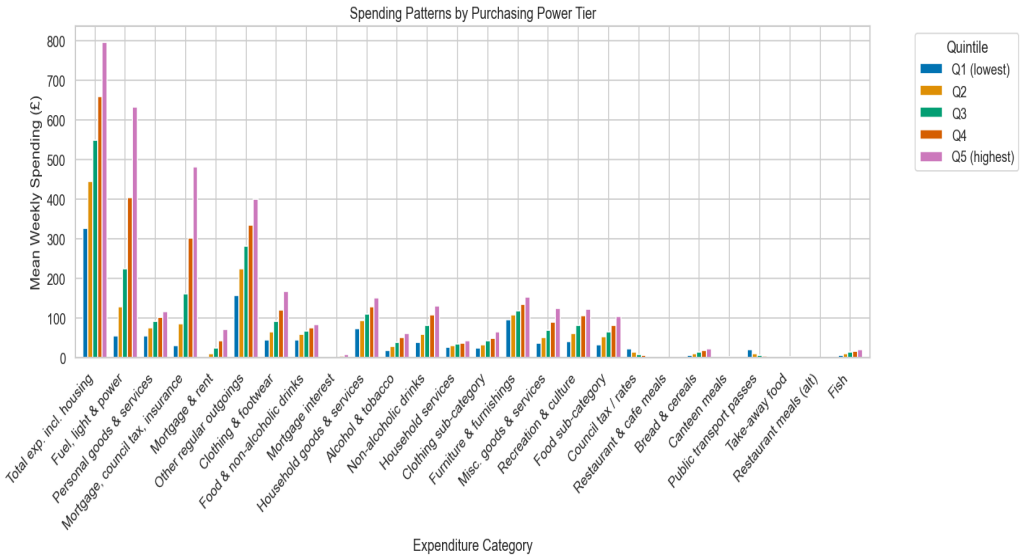

Figure 4: Mean weekly spending by purchasing power tier across expenditure categories.

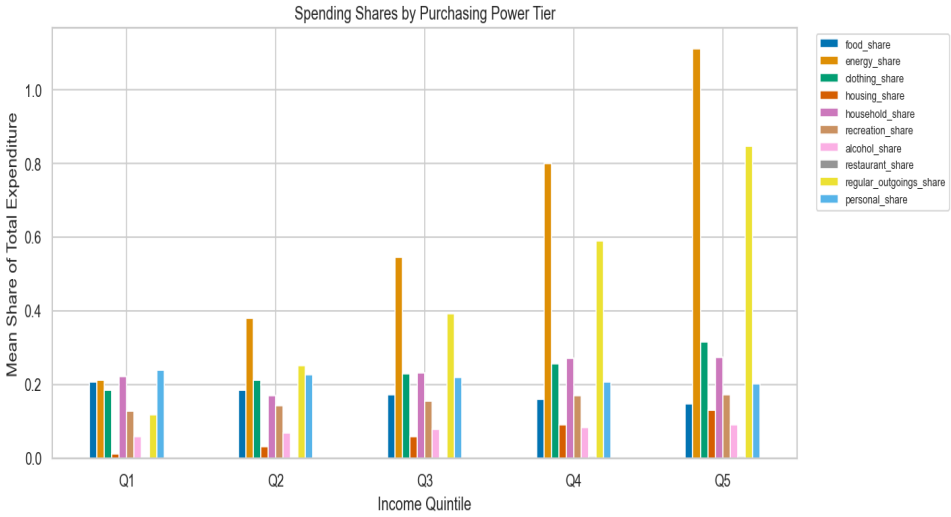

Figure 5: Spending share ratios by purchasing power tier, illustrating Engel's Law pattern.

3. Data Preparation

A temporal split was adopted for this analysis, with 2021–2022 data allocated to training and validation (8,578 + 1,514 households) and the entire 2023 survey reserved for testing (4,202 households). This design forces the model to generalise across spending patterns shaped by the cost-of-living crisis. Within the training period, the validation set is drawn by stratified sampling to preserve class balance.

The preprocessing pipeline uses scikit-learn's ColumnTransformer (Pedregosa *et al.*, 2011) for reproducibility. For the 38 continuous variables (28 expenditure and demographic measures plus 10 engineered spending-share ratios), a custom CoerceNumeric transformer addressed empty strings that disrupted sklearn's SimpleImputer, followed by median imputation and standard scaling. The 24 categorical variables received mode imputation and one-hot encoding, with unseen categories at test time handled gracefully, producing 204 preprocessed features in total.

Ten spending-share ratios (such as the ratio of food expenditure to total expenditure, energy to total, and housing to total) were additionally engineered, motivated by Engel's Law: lower-income households allocate a higher proportion of spending to necessities, even where absolute expenditure amounts are comparable. Among these ratios, housing share exhibits the strongest Q5/Q1 differential (12.9x), followed by regular outgoings share (7.3x).

To further focus the model on the strongest signals, a preliminary Gradient Boosting classifier ranked all 204 features by importance. The top 35 were retained, led by mortgage, council tax, and insurance (importance = 0.354). This reduction in dimensionality from 204 to 35 features mitigates overfitting while preserving the most informative predictors.

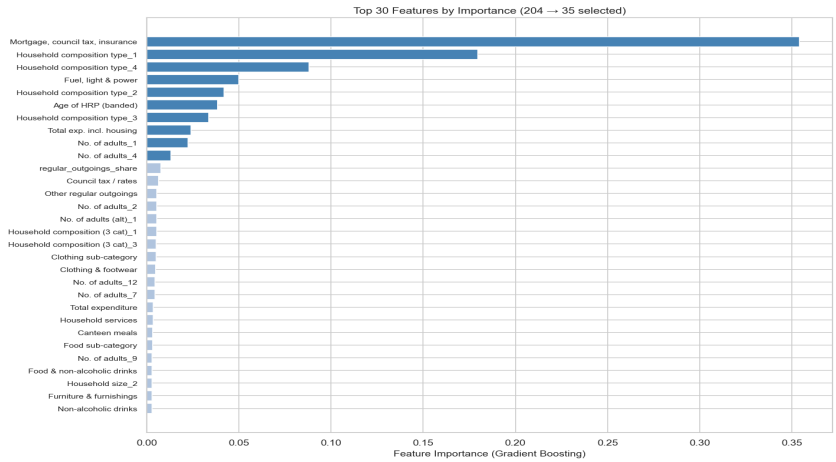

Figure 6: Feature importance ranking used for selection (204 to 35 features).

Regional analysis further contextualises these patterns. Box plots of equivalised income by government office region reveal that London and the South East exhibit the highest purchasing power, while the North East and Wales have the lowest. This geographic disparity reinforces the policy case for a spending-based classifier that could facilitate regionally targeted welfare interventions without income disclosure.

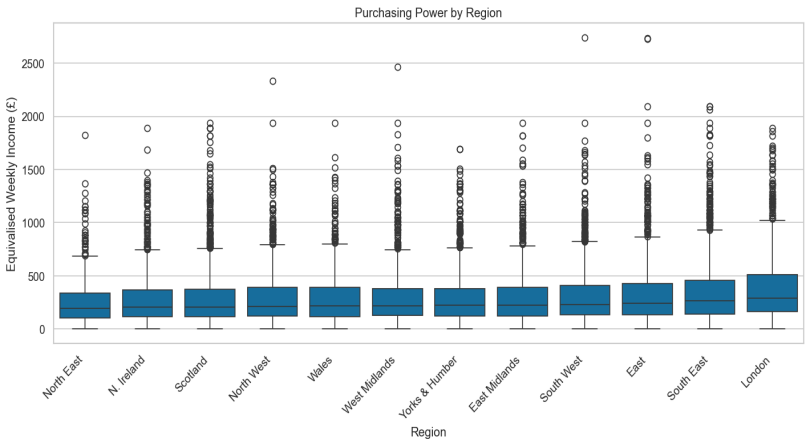

Figure 7: Purchasing power distribution by UK government office region.

To ensure pipeline integrity, eighteen automated pipeline validation tests were implemented, all of which passed. These verify that no NaN values survive preprocessing, no households appear in both training and test sets, class proportions are preserved across splits, and no income-derived leakage variable remains in the feature set.

4. Model Exploration

The exploratory analysis revealed non-linear spending patterns, mixed feature types, and overlapping middle quintiles, characteristics that favour flexible, ensemble-based learners over simple linear models. Six architectures were compared using five-fold stratified cross-validation alongside held-out validation:

Model	CV F1 (±SD)	Val F1	Val Acc	Role
Neural Network	0.7001 ± 0.007	0.7007	0.6988	Deep learning
Gradient Boosting	0.6802 ± 0.005	0.6722	0.6711	Ensemble baseline
SVM (RBF)	0.6713 ± 0.011	0.6762	N/A	Non-linear comparison
Random Forest	0.6642 ± 0.005	0.6594	0.6579	Tree ensemble
Logistic Regression	0.6520 ± 0.014	0.6555	0.6585	Linear baseline
PCA + LogReg	0.3838 ± 0.010	0.3936	0.4108	Dim. reduction

Table 1: Model comparison by cross-validation and validation macro F1.

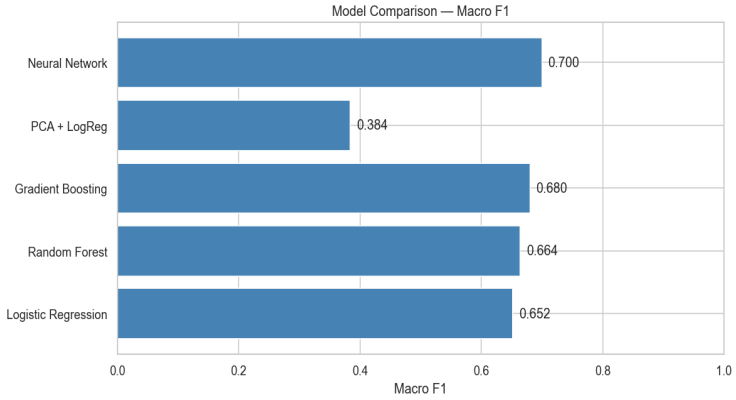

Figure 8: Model comparison by macro F1 score. Neural Network leads all models.

A clear hierarchy emerges. PCA + Logistic Regression collapses to F1 = 0.39, confirming that compressing 38 features into 22 principal components discards the fine-grained expenditure detail needed for quintile separation and that categorical variables, excluded from PCA, carry essential signal. Logistic Regression (F1 = 0.66) provides the lower bound, while the tree ensembles and SVM cluster between 0.66 and 0.68. SVM (Val F1 = 0.68) outperformed both Logistic Regression and Random Forest, lending empirical support to the hypothesis that non-linear models are better suited to this task. However, SVM was not selected for the final pipeline owing to its lack of feature importance measures and native probability outputs.

The Neural Network (64 to 32 units, ReLU, dropout 0.4/0.3) achieved the highest validation performance at F1 = 0.7007 (Chollet *et al.*, 2015). Trained with sparse categorical cross-entropy and early stopping (patience = 10), it manages the bias-variance trade-off more effectively than the tree ensembles (Geron, 2022).

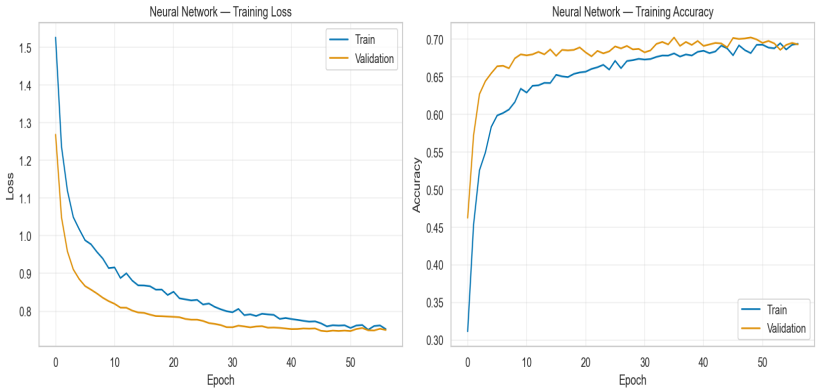

Figure 9: Neural Network training curves — loss (left) and accuracy (right). Early stopping triggers after convergence.

Learning curves for the tree ensembles revealed substantial train-validation gaps, which necessitated three corrective measures: tighter regularisation constraints, feature selection (reducing the feature space from 204 to 35), and the introduction of early stopping.

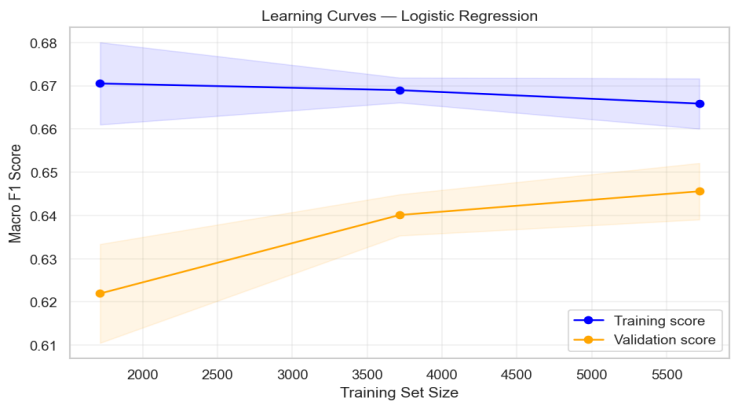

Figure 10: Gradient Boosting learning curves showing train-validation gap.

A notable agent error was identified during hyperparameter tuning, where the agent set the tuning objective to accuracy rather than macro F1. Although the numerical impact is small with balanced classes, the conceptual error is significant, as accuracy disproportionately rewards correct classification of the majority class, whereas macro F1 assigns equal weight to all quintiles. This was identified during code review and corrected before final evaluation.

5. Fine-Tuning and Evaluation

Hyperparameter tuning was applied to all scikit-learn models (Geron, 2022): Logistic Regression via grid search (optimal regularisation strength C = 50.0), Random Forest via randomised search (optimal depth of 20 with balanced subsample weighting), and Gradient Boosting via randomised search (learning rate of 0.05, depth of 4, 500 estimators, 70% subsampling). Tuning improved validation F1 for all three models (Gradient Boosting from 0.6722 to 0.6798, Random Forest from 0.6594 to 0.6777, and Logistic Regression from 0.6555 to 0.6592), yet none surpassed the Neural Network.

On the held-out 2023 test set, the Neural Network achieves a macro F1 of 0.6921 and accuracy of 0.6959. The modest gap from validation (0.7007) to test (0.6921) reflects the temporal shift in spending behaviour under continued cost-of-living pressures, consistent with the distributional drift that a temporal split is designed to capture.

Quintile	Precision	Recall	F1	Support
Q1 (lowest)	0.78	0.81	0.80	838
Q2	0.58	0.63	0.61	784
Q3	0.58	0.61	0.59	834
Q4	0.66	0.56	0.61	841
Q5 (highest)	0.86	0.85	0.85	905

Table 2: Per-quintile classification performance on the 2023 test set.

The per-quintile breakdown illuminates the model's principal strengths and limitations. The extremes are well-separated: Q1 (F1 = 0.80) and Q5 (F1 = 0.85) have distinctive spending fingerprints that the model identifies with high confidence. By contrast, the middle quintiles prove more difficult to separate; Q3 achieves only F1 = 0.59, reflecting the economic reality that middle-income households genuinely spend in similar patterns. Crucially, 1,166 of 1,278 misclassifications (91.2%) fall between adjacent quintiles; the model virtually never confuses the poorest with the wealthiest, respecting the ordinal structure of income.

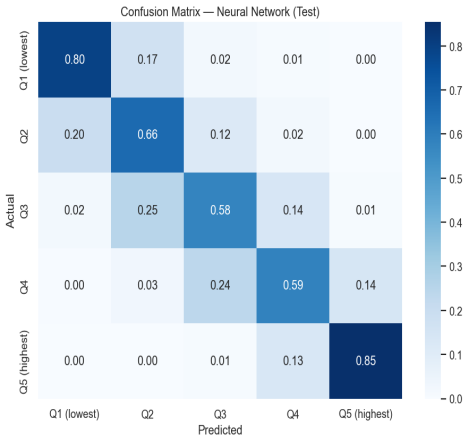

Figure 11: Normalised confusion matrix (Neural Network, test set). Errors concentrate on the diagonal's neighbours.

To assess whether the model's predicted probabilities are trustworthy, temperature scaling (Guo *et al.*, 2017) was applied as a post-hoc calibration method. The optimal temperature T = 0.97, learned by minimising negative log-likelihood on the validation set, indicates slight underconfidence; scaling therefore sharpens rather than softens predictions.

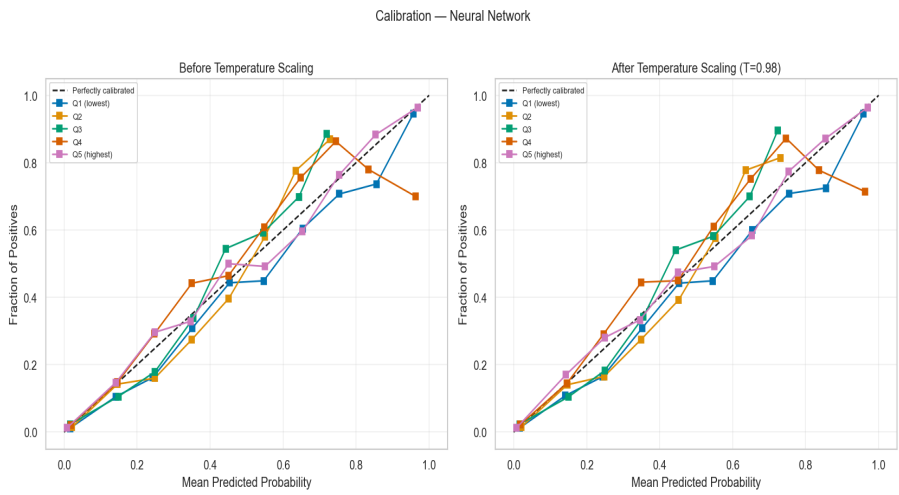

Figure 12: Neural Network calibration curves before (left) and after (right) temperature scaling (T = 0.97).

To ground the key design choices in evidence rather than narrative, three controlled ablation experiments were conducted, each altering exactly one factor from the final pipeline.

Experiment	Val F1	Test F1	Note
Final pipeline (baseline)	0.7007	0.6921	Top 35 features, temporal split
All features (no selection)	0.6995	0.6977	All 204 preprocessed features
Random split	0.7431	0.7141	Random stratified split
With leakage (anon_income)	0.9279	0.9333	Demonstrates cheating

Table 3: Ablation study validating key design decisions.

The temporal split imposes a meaningfully harder test than random splitting (F1 drops from 0.7141 to 0.6921), confirming it captures real distributional drift. Feature selection yields a marginal trade-off, suggesting the top 35 features carry most of the signal. Including the income variable itself inflates F1 to 0.93, confirming that leakage exclusion is essential and that the model's genuine performance is not an artefact of data leakage. Slight differences from the main results reflect neural network retraining variance; the relative comparisons remain informative.

6. Final Solution

The Neural Network (64 to 32 units, dropout 0.4/0.3, early stopping) is selected as the final model, correctly classifying 70% of households into the appropriate purchasing-power quintile, with 91.2% of errors falling between adjacent tiers. The narrow 0.9-point gap between validation and test F1 provides confidence that the model generalises reliably to unseen time periods.

Field	Details
Purpose	Predict purchasing power tier from spending patterns
Not for	Individual financial decisions; real-time inference; non-UK data
Data	UK LCFS 2021–2023, 14,294 households (ONS via UK Data Service)
Features	35 selected from 62 candidates (52 original + 10 engineered) via importance ranking; no income inputs
Test performance	Macro F1 = 0.6921, Accuracy = 0.6959
Key limitation	Middle-quintile confusion (Q3 F1 = 0.59)

Table 4: Model card summary.

Several limitations warrant acknowledgement. First, middle quintiles share genuinely similar spending patterns, placing a fundamental ceiling on separability. Second, column a116 (age of household reference person) consisted entirely of empty strings across all survey years, resulting in the loss of a potentially informative demographic predictor. Third, survey weights were not incorporated, meaning the analysis assumes simple random sampling. Fourth, self-

reported expenditure is inherently subject to measurement error from under- and over-reporting. Fifth, three years of survey data may not capture longer-term economic trends or structural shifts in household spending. Finally, the OECD modified equivalence scale is itself a simplification that does not fully account for economies of scale in larger households.

Reflecting on the role of agent tooling throughout this project, Claude Code accelerated development substantially by scaffolding the pipeline, proposing architectures, and generating tests, yet it required continuous human oversight. The accuracy-as-objective mistake during tuning exemplifies how AI tools can introduce subtle methodological errors that pass unnoticed without domain understanding. The agent also initially framed the SVM experiment as a clear rejection despite its outperforming two other models; honest reporting of results required manual correction. The lesson is consistent: the agent is a powerful accelerator, not an autonomous analyst.

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